



Introduction to business logic 15m

In this section, you will learn about the business logic layer in PHP. Before this, we'll give a summary of the PHP layers' responsibilities and how they interact.

There are three layers: the presentation layer, the business logic layer and the data access layer.

- **Presentation layer** – This is the layer at which users interact with the application and the output will be presented to the users on this layer. It acts as an interface between the user and the application.
- **Business logic layer** – This acts as an intermediate layer between the presentation and the data access layer. It is responsible for performing the calculations written as per the business rules.
- **Data access layer** – The layer at which the data is managed.

All three layers play an important role in building an application.

In an application, when a user is accessing it or requesting information (by sending a query), they are using the presentation layer (the user interface). The business logic layer then assists the user in receiving a response to their query by transferring their request to the data access layer. The data access layer then processes the query and performs calculations at the business logic layer, which then returns a response to the presentation layer to be presented to the user.

Take, for example, a health insurance company website where customers can view their health insurance policies. The customers use a web browser to visit the website, and the browser interacts with the application's presentation layer code.

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The data layer will be responsible for the customer policy data being saved and maintained in a database. The application code determines which policies a customer is able to view, and any calculations to be performed are done in the business logic layer.

What is the business logic layer?

The business logic layer sits between the presentation layer and the data access layer. Its job is to manage the business rules, logic and calculations needed for the application to work. This layer is sometimes called the domain layer because it deals with both business and domain-specific logic.

In simple terms, this layer decides how data should be used, how information from the database can be processed, what calculations need to be done and what actions are allowed or not allowed within the application.

Watch *What is Business Logic in Software Development?* (Be a Better Dev, 2020) (5:03) to understand the business logic layer. This video gives two practical examples to demonstrate how the business logic layer works in real life.



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What is Business Logic in Software Development?

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Advantages of the business logic layer

The advantages of the business logic layer are:

- **Code maintenance can become easier.** Maintaining the code will be easy when we are using the business logic layer as it supports multitier architecture. The layering provides a separation of concerns, contextualising code maintenance.
- **Security.** This architecture provides security to data as it is obvious that the presentation layer doesn't directly interact with the data access layer. Therefore, it prevents any kind of data loss and ensures that the data is secure at the data layer.
- **Application releases.** It makes application release rollout easy. Because the business logic layer is the only one to be updated every time, we do not need to alter the other layers of architecture; that is, the presentation layer and the data access layer.
- **Ease of learning.** Learning is straightforward since you only need to focus on either the presentation, data, or business layer. This allows for quicker mastery of relevant

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parts and efficient development as all layers can operate simultaneously.

Disadvantages of the business logic layer

- **Expensive.** It can be difficult and expensive to install as well as maintain this layer in databases.
- **Source control** is very difficult to perform properly with existing procedures.
- It makes it difficult to use the code again and again, **decreasing code reusability**.



Thoughts on this page?

